

PIK3CA variants selectively initiate brain hyperactivity during gliomagenesis

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Glioblastoma is a universally lethal form of brain cancer that exhibits an array of pathophysiological phenotypes, many of which are mediated by interactions with the neuronal microenvironment^{1,2}. Recent studies have shown that increases in neuronal activity have an important role in the proliferation and progression of glioblastoma^{3,4}. Whether there is reciprocal crosstalk between glioblastoma and neurons remains poorly defined, as the mechanisms that underlie how these tumours remodel the neuronal milieu towards increased activity are unknown. Here, using a native mouse model of glioblastoma, we develop a high-throughput in vivo screening platform and discover several driver variants of PIK3CA. We show that tumours driven by these variants have divergent molecular properties that manifest in selective initiation of brain hyperexcitability and remodelling of the synaptic constituency. Furthermore, secreted members of the glypican (GPC) family are selectively expressed in these tumours, and GPC3 drives gliomagenesis and hyperexcitability. Together, our studies illustrate the importance of functionally interrogating diverse tumour phenotypes driven by individual, yet related, variants and reveal how glioblastoma alters the neuronal microenvironment.

In the pursuit of cataloguing the spectrum of somatic mutations for all tumour types, cancer genomics is rapidly approaching a state of saturation mutagenesis⁵. These crucial advances promise to usher in a new era of personalized medicine, in which patient-specific genomic information is applied in the clinic. One major obstacle in achieving this goal is decoding driver and passenger mutations from these cohorts, as not all mutations affect malignancy, whereas others exert context-specific functions. Mutation ‘hotspots’ represent variants in key driver genes that occur with high frequency and are found across different types of cancer⁶. Although widely accepted as the convention for prospective selection of driver mutations, this approach overlooks most mutations and, crucially, does not account for how cellular context influences variant function^{7–12}. Whether related variants exert differential effects on both tumour and microenvironmental phenotypes remains unknown. These limitations illustrate the need for high-throughput, functional and phenotypic screening of individual variant cohorts in contextually appropriate in vivo systems.

Identifying PIK3CA driver variants

The RTK–RAS–PI3K pathway is a key driver of tumorigenesis across all cancers, with 90% of glioblastoma tumours exhibiting alterations in this pathway^{13,14}. Among the specific genes in this pathway, mutations in the

PI3K catalytic subunit *PIK3CA* are found in 11% of glioblastoma tumours¹³. Sequencing of glioblastoma samples revealed several known hotspot mutations that drive tumorigenesis in several cancer lineages (E545K and H1047R), as well as a series of 63 in-frame mutations (as indexed in the Catalogue Of Somatic Mutations In Cancer (COSMIC)) that remain largely unclassified (Fig. 1a, Supplementary Table 1). To decode which of these PIK3CA variants function as drivers of glioblastoma, we established an in vivo complementation screening platform for glioblastoma. Our mouse glioblastoma model relies on in utero electroporation (IUE) and CRISPR–Cas9-mediated knockout of *Nf1*, *Trp53* (also known as *p53*) and *Pten* (termed ‘3xCr’), with 50% of the mice succumbing to tumours by postnatal day (P) 84¹⁵ (Fig. 1d, black line, Extended Data Table 1). Removing the *Pten* guide RNA (termed ‘2xCr’) extends median survival to P203¹⁶, providing ample dynamic range to screen for factors that complement the loss of PTEN (Fig. 1d, grey line, Extended Data Table 1). Because PTEN catalyses the reverse reaction of PIK3CA—phosphorylation of phosphatidylinositol-4,5-bisphosphate (PtdIns(4,5)P₂, or PIP₂) to phosphatidylinositol-3,4,5-trisphosphate (PtdIns(3,4,5)P₃, or PIP₃)—we complemented *Pten* loss with overexpression of E545K or H1047R—two bona fide hotspot driver mutations of *PIK3CA*¹². Both of these variants accelerated tumour growth, which demonstrates that known PIK3CA drivers complement PTEN loss in our system (Fig. 1d, red and green lines, Extended Data Fig. 1a, b, Extended Data Table 1).

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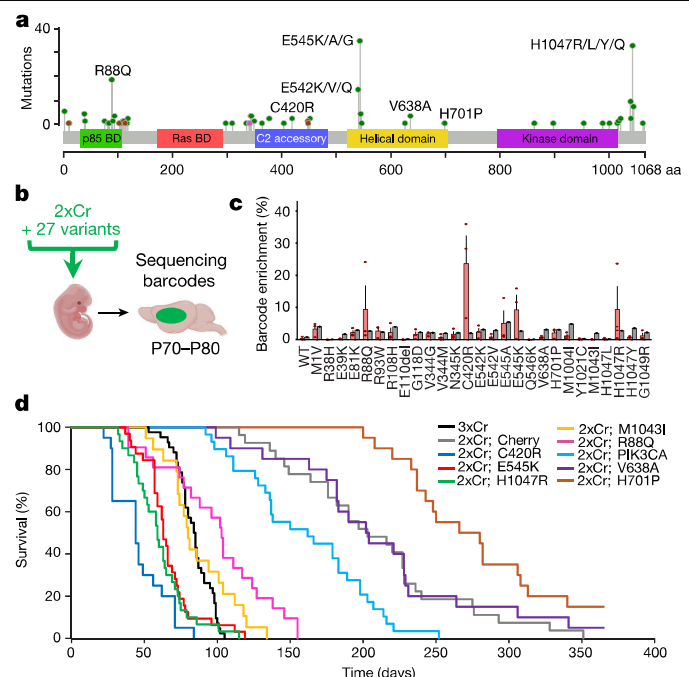


Fig. 1 | In vivo screening identifies novel driver PIK3CA variants in glioma.

a, Glioblastoma-associated PIK3CA mutations indexed in COSMIC (by June 2019): missense (green), in-frame deletions and insertions (brown), or nonsense (pink) (Supplementary Table 1). **b**, Schematic of in vivo competition assay in which *Nf1* and *Trp53* are deleted (2xCr) and 27 uniquely barcoded *Pik3ca* alleles are co-electroporated. **c**, Next-generation sequencing for barcode amplification, showing the barcode for each allele (red) and the input signal (black). $n = 3$ tumours. Data are mean and s.e.m. **d**, Kaplan-Meier curve of all tested variants. $n = 42$ (3xCr); $n = 28$ (Cherry); $n = 20$ (C420R); $n = 32$ (E545K); $n = 30$ (H1047R); $n = 19$ (M1043I); $n = 21$ (R88Q); $n = 29$ (WT); $n = 20$ (V638A); $n = 20$ (H701P). See Extended Data Table 1 for full survival statistics.

Our finding that established *PIK3CA* hotspots can drive in vivo tumorigenesis led us to screen a cohort of PIK3CA variants found in human glioblastoma (Fig. 1a, Supplementary Table 1). To achieve this, we used DNA barcoding together with a pooled screening strategy^{17,18} (Methods) in our sensitized 2xCr model (Fig. 1b, Extended Data Fig. 2). After glioma formation, we used targeted sequencing of tumours for barcodes as a surrogate for variant enrichment within the tumours. This analysis demonstrated that the established hotspots (E545K and H1047R) are enriched along with other variants (R88Q and C420R) that have not been characterized in glioma (Fig. 1c). To confirm driver function of these variants, we individually expressed them in the 2xCr model, finding that they all complement the loss of PTEN and drive tumorigenesis, while also pathologically resembling glioblastoma (Fig. 1d, Extended Data Figs. 1a, b, 3, Extended Data Table 1). These data indicate that our approach can identify established hotspots, as well as new PIK3CA driver variants in glioma.

PIK3CA tumours exhibit synaptic profiles

Although the tested driver variants accelerated tumour-associated death compared to the wild-type PIK3CA control, they did so to varying degrees (Fig. 1d). To understand a molecular basis for differences in tumorigenic potential between these variants, we performed reverse-phase protein array (RPPA) analysis on lysates collected from tumours driven by each variant, finding differential activation of PI3K pathway components (Fig. 2a). C420R and H1047R demonstrated the highest PI3K activity and were also the strongest drivers of tumorigenesis (Fig. 1d), whereas E545K, R88Q and M1043I exhibited less activation. These differences in PI3K tone and survival suggest that individual

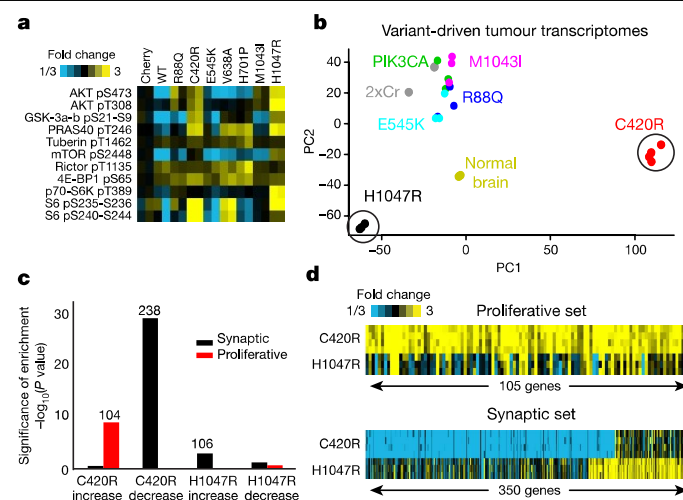


Fig. 2 | Tumours driven by PIK3CA variants exhibit diverse molecular properties.

a, Tumour lysate RPPA for proteins of the PI3K-AKT-MTOR pathway¹²; $n = 2$ per variant. **b**, Principal component (PC) analysis against top 2,000 variable genes across variant tumours from RNA-seq data. $n = 2$ (normal brain); $n = 4$ (C420R); all other variants $n = 3$. **c**, Enrichment analysis of the top genes increased or decreased in C420R and H1047R variants ($P < 0.01$ for both wild-type and Cherry control comparisons, by linear model) for gene sets associated with synapse function or proliferation. Number above the bars represent genes from each set involved in observed significant overlap. P values were determined by one-sided Fisher's exact test; $n = 4$ (C420R); $n = 3$ (H1047R). **d**, Differential gene expression patterns (relative to average of wild-type and Cherry controls) for tumours driven by C420R or H1047R for proliferative or synaptic genes.

variants promote tumorigenesis through distinct mechanisms, which we examined via RNA sequencing (RNA-seq) analysis of end-stage tumours driven by these variants. Principal component analysis of the transcriptomes revealed that tumours driven by E545K, M1043I, R88Q, wild-type PIK3CA and the 2xCr control are highly correlated (Fig. 2b). Notably, principal component analysis revealed that the transcriptomes of tumours driven by H1047R and C420R are vastly different from this core set of variants and from one another (Fig. 2b, circles). Focusing on the molecular differences manifest in H1047R- and C420R-driven tumours, we performed Gene Ontology analysis of the transcriptome data, and found that genes associated with proliferation and synapses are differentially expressed (Fig. 2c, Supplementary Table 2). Although C420R tumours demonstrated a notable increase in proliferative genes, we found two distinct patterns of synaptic gene dysregulation, with C420R tumours showing suppression of one subgroup of synaptic genes and H1047R tumours revealing robust upregulation of an entirely different set of synaptic genes (Fig. 2d). Together, these molecular analyses reveal that gliomas driven by different PIK3CA variants are endowed with distinct molecular features that can be uncoupled from PI3K activation (that is, C420R and H1047R). Moreover, these findings highlight how driver variants endowed with single amino acid differences, exhibiting seemingly similar phenotypic effects, can have distinct underlying molecular properties.

Selective induction of hyperexcitability

Given the distinct molecular features of tumours driven by C420R and H1047R mutations, we next determined whether these tumours exhibit unique biological properties. Because tumours driven by C420R display a pronounced increase in genes associated with proliferation (Fig. 2c, d) and cause a shorter median survival time in mice when compared to H1047R-driven tumours (44 versus 59 days), we evaluated tumour growth and proliferation during early tumorigenesis. Using BrdU labelling and

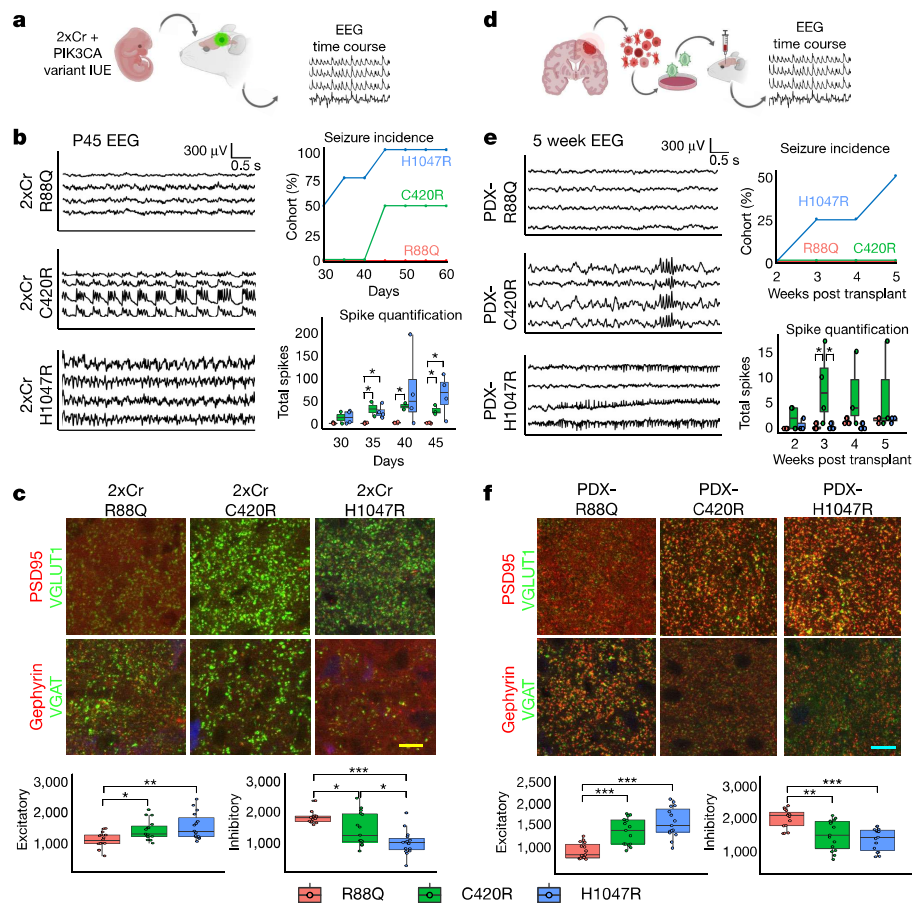


Fig. 3 | C420R and H1047R tumours promote hyperexcitability and synaptic imbalance across tumour models. **a**, Schematic of experimental plan for IUE model. **b**, EEG traces of select variant tumour mice at P45 and quantification of EEG analyses. Traces for C420R and H1047R occurred during seizures. Seizure incidence (top) and quantification of spike activity (bottom) are shown; $n = 4$ mice per variant. Data in seizure incidence graph denote mean values. Box plots denote the median value, interquartile range (25th–75th percentiles), with whiskers extending to 1.5 times the interquartile range. **c**, Antibody staining of excitatory and inhibitory synapses in P30 mouse brains at peritumoral margins (top) and quantification from a $34,000\text{-}\mu\text{m}^2$ field of view (bottom). GFP signal (blue, not labelled) denotes tumour. $n = 3$ mice; $n = 5$

technical replicates. Box plots as in **b**, **d**, Schematic of PDX model experiments. **e**, EEG traces (left) and quantification (right) of mice bearing PDX tumours expressing PIK3CA variants five weeks after transplantation. Trace for PDX-C420R demonstrates aberrant EEG patterns without associated seizures; trace for PDX-H1047R was during an electrographic, non-convulsive seizure. $n = 4$ mice per variant. Box plots are as in **b**, **f**, Antibody staining of excitatory and inhibitory synapses from peritumoral margins of PDX brains (top) and quantification from a $34,000\text{-}\mu\text{m}^2$ field of view (bottom). $n = 3$ mice; $n = 5$ technical repeats. Box plots as in **b**, **f**. Scale bars, $12.5\text{ }\mu\text{m}$ (**c**) and $10\text{ }\mu\text{m}$ (**f**). $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, one-way analysis of variance (ANOVA).

MRI imaging (Extended Data Fig. 1c–e), we found that C420R-driven tumours exhibit an enhanced proliferative phenotype compared to tumours driven by H1047R and R88Q, thus corroborating the underlying molecular features of these variants. The other defining molecular feature of C420R- and H1047R-driven tumours is dysregulation of distinct cohorts of synapse-associated genes (Fig. 2c, d). It is well-established that synaptic imbalance can result in extensive changes in network excitability, which in some cases culminates in seizure activity^{19,20}. Crucially, seizures are an early pathophysiological hallmark of malignant glioma that are recapitulated in our mouse model of glioma^{15,21–23}. Therefore, we determined whether mice bearing tumours driven by C420R and H1047R exhibit pronounced network hyperexcitability during the early stages of tumour progression (Fig. 3a). We performed serial video electroencephalography (EEG) recordings every 5 days during the P30–P60 interval (Extended Data Fig. 4), and found that mice bearing tumours driven by H1047R show an earlier onset of seizures (50% of the cohort exhibited convulsive seizures at P30) than mice bearing tumours driven by C420R (seizures first appear at P45) (Fig. 3b). EEG recordings of these mice during this interval revealed considerable increases in network hyperexcitability, with both groups of mice demonstrating similar increases

in spiking frequency as early as P30–P35 (Fig. 3b). By contrast, mice bearing R88Q-driven tumours did not demonstrate convulsive seizures or changes in EEG activity during the P30–P60 interval (Fig. 3b). These variant-specific increases in hyperexcitability correlate with the relative changes in synaptic gene cohorts of C420R and H1047R tumours, which suggests that tumours driven by these variants result in early synaptic imbalance during the formative stages of tumour progression.

To determine whether C420R and H1047R tumours exhibit changes in synaptic phenotypes in the tumour microenvironment during early tumorigenesis, we isolated C420R- and H1047R-driven tumours from P30 mice and stained for makers of excitatory (VGLUT1 and PSD95) and inhibitory (VGAT and gephyrin) synapses at the peritumoral margins (Extended Data Fig. 5). These studies revealed significant increases in excitatory synapses at the peritumoral margins of both C420R and H1047R tumours compared with R88Q tumours (Fig. 3c). Notably, analysis of inhibitory synapses revealed a decrease in their numbers surrounding C420R and H1047R tumours, compared to R88Q (Fig. 3c). These changes in synaptic constituency—increases in excitatory and decreases in inhibitory synapses—in neurons adjacent to C420R- and H1047R-driven tumours further indicate that the observed increases

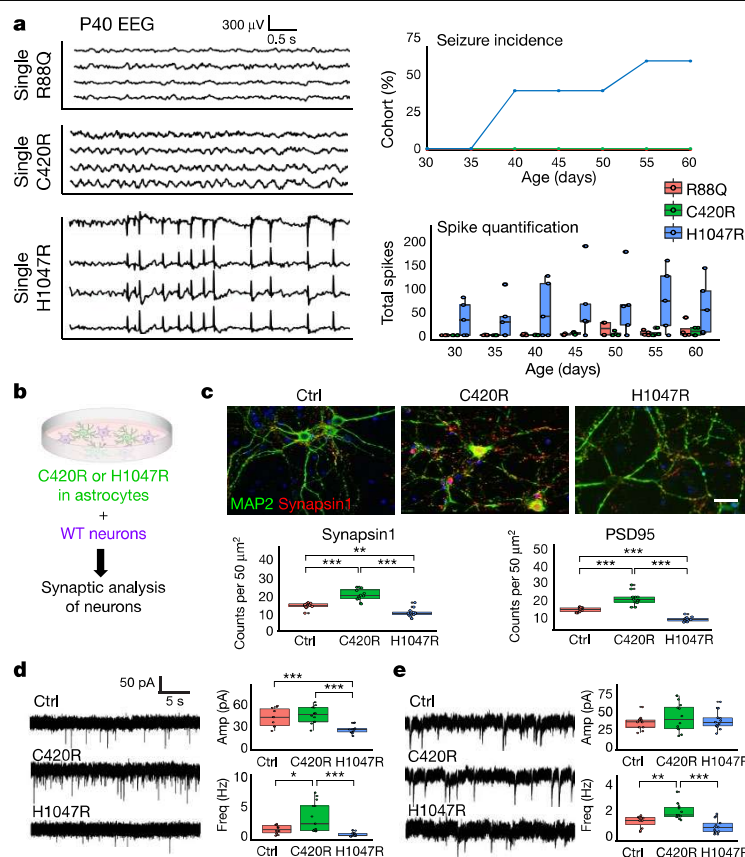


Fig. 4 | C420R and H1047R differentially promote synaptic imbalance. **a**, EEG traces and quantification of variants overexpressed in mouse brains at P40. Trace for H1047R was during seizure. Seizure incidence and quantification of total spike activity are shown; $n = 4$ mice per variant. Box plots are as in Fig. 3b. No significant ($P < 0.05$) P values were calculated by one-way ANOVA. **b**, Schematic of astrocyte–neuron co-culture. WT, wild type. **c**, Top, antibody staining for synapsin1 (red) and MAP2 (green) of astrocyte–neuron co-cultures; astrocytes infected with control (ctrl) or C420R or H1047R viruses. Bottom, quantification of PSD95 and synapsin1 staining. $n = 12$ technical replicates per condition. Data are mean and s.e.m. Scale bar, 50 μ m. **d**, **e**, Representative traces (left) and quantification (right) of excitatory (**d**) or inhibitory (**e**) postsynaptic currents from whole-cell recording of neurons co-cultured on astrocytes infected with control, or C420R or H1047R viruses. $n = 12$ technical replicates per condition. Data are mean and s.e.m. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, one-tailed independent t -test; Tukey’s test was used to compare individual mean values.

in network hyperexcitability result, in part, from synaptic imbalance. Next, we extended these functional studies to patient-derived xenograft models (PDX), by generating glioma stem-cell lines²⁴ that individually overexpress H1047R, C420R and R88Q (Fig. 3d). These lines were subsequently transplanted into the brains of SCID mice, in which we assessed the onset of hyperexcitability via EEG, and found that mice bearing PDX-H1047R exhibited early onset of seizures whereas mice bearing PDX-C420R exhibited early onset of network hyperexcitability (Fig. 3e). PDX-R88Q did not demonstrate early onset of either seizures or hyperexcitability (Fig. 3e). Moreover, cellular analysis of the peritumoral synaptic constituency revealed that both PDX-H1047R and PDX-C420R tumours provoke an increase in excitatory synapses, coupled with a decrease in inhibitory synapses compared to PDX-R88Q tumours (Fig. 3f). These alterations in synaptic constituency are met with congruent changes in cell proliferation, as PDX-H1047R and PDX-C420R exhibit increased incorporation of BrdU (Extended Data Fig. 1f). Together, these functional studies in PDX models reinforce our conclusion that these variants selectively promote changes in network hyperexcitability and synaptic constituency.

Variant-specific synaptogenic mechanisms

The synaptic changes at the peritumoral margin in both the native and PDX models suggest that these variants function via

cell-non-autonomous mechanisms. However, several PIK3CA variants are associated with monogenic paediatric overgrowth syndromes and can promote hyperexcitability on their own, including H1047R, which suggests that they may also function cell autonomously^{25–27}. To address this, we performed individual IUE, outside the tumorigenic context, with H1047R, C420R and R88Q, in conjunction with serial EEG recordings every 5 days during the P30–P60 interval (Extended Data Fig. 4). Mice expressing H1047R exhibited increased hyperexcitability as early as P30, coupled with seizures at P40, whereas mice expressing C420R and R88Q did not demonstrate any changes during this period (Fig. 4a); we did not observe any changes in cell proliferation (Extended Data Fig. 1g–i). These data indicate that H1047R promotes hyperexcitability on its own, whereas C420R cannot—which suggests that these variants use different mechanisms to alter synaptic constituency. Next, we used an established neuron–glial co-culture system¹⁵, which enables us to directly assess the cell-non-autonomous contributions of these variants to synaptogenesis (Fig. 4b). We overexpressed C420R and H1047R in mouse astrocytes co-cultured with neurons, and found that C420R promotes synapse formation between neurons as measured by the expression of synaptic markers (Fig. 4c). Whole-cell recordings revealed increased frequency of excitatory and inhibitory postsynaptic currents (EPSCs and IPSCs, respectively) in neurons when co-cultured with astrocytes expressing C420R (Fig. 4d, e). By contrast, overexpression of H1047R in astrocytes did not enhance synapse formation on neurons

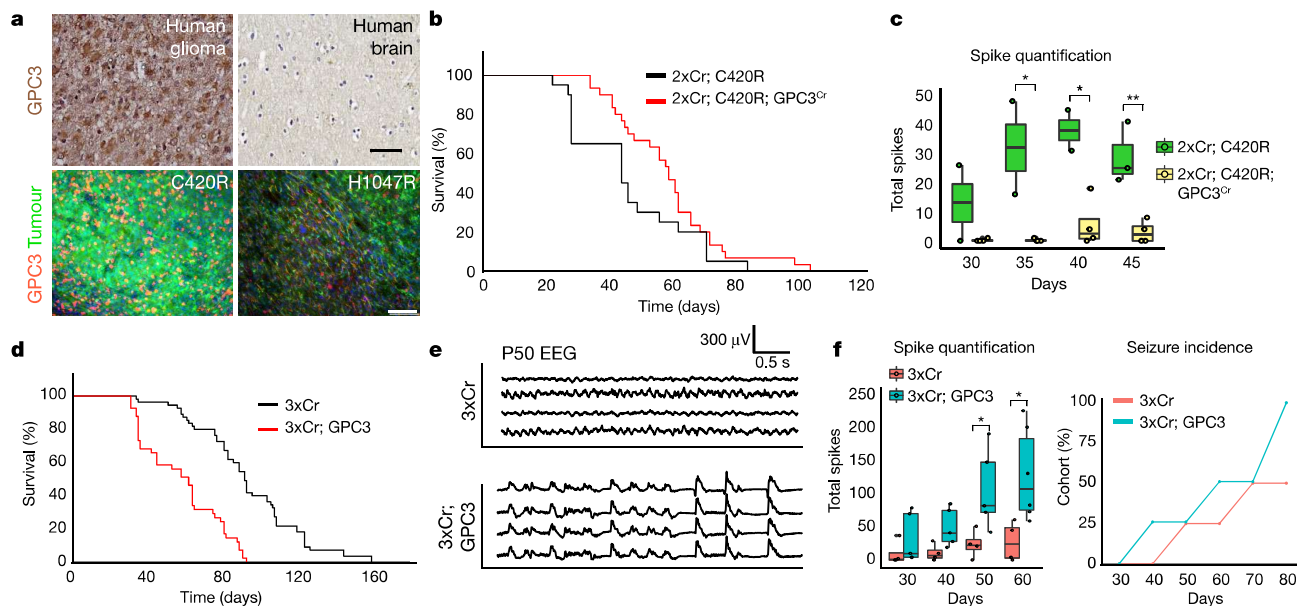


Fig. 5 | GPC3 promotes gliomagenesis and synaptic imbalance. **a**, Top, GPC3 staining of human glioblastoma and healthy human brain control. Staining was not independently repeated. Bottom, GPC3 antibody staining of C420R and H1047R tumours from P30 mouse brains. $n = 4$ mice for each variant. These experiments were independently repeated four times. Scale bars, 50 μm (top) and 100 μm (bottom). **b**, Kaplan–Meier curve comparing GPC3 deletion (GPC3^{cr}) in 2xCr; C420R tumours. $n = 20$ (C420R) and $n = 30$ (C420R; GPC3^{cr}) mice. Statistics are in Extended Data Fig. 8b. **c**, Quantification of total spike activity shows that EEG spike activity decreases with GPC3 deletion. $n = 4$ mice

per variant. Box plots are as in Fig. 3b. $*P < 0.05$, $**P < 0.01$, one-way ANOVA. $n = 4$ mice each. **d**, Kaplan–Meier curve comparing *Gpc3* overexpression in 3xCr tumours. $n = 55$ (3xCr) and $n = 41$ (3xCr; GPC3) mice. Statistics are in Extended Data Fig. 8b. **e**, EEG traces of *Gpc3* overexpression in 3xCr tumours at P50. **f**, Quantification of total spike activity (left) and seizure incidence (right). $n = 4$ mice (3xCr) and $n = 5$ mice (3xCr; GPC3) for spike quantifications; $n = 4$ mice per condition for seizure incidence. Box plots are as in Fig. 3b. $*P < 0.05$, one-way ANOVA.

(Fig. 4d, e). Together, these results illustrate mechanistic differences between H1047R and C420R. H1047R promotes brain hyperexcitability on its own, through cell-autonomous mechanisms, whereas C420R promotes brain hyperexcitability specifically in the context of a glial tumour, through cell-non-autonomous mechanisms.

GPC3 promotes tumorigenesis

To understand how C420R exerts a cell-non-autonomous effect on synaptogenesis, we further analysed the RNA-seq data from the variant-driven tumours, focusing on a set of genes that is secreted from glia and promotes synapse formation onto neurons²⁸. Among this group, we found that members of the glypican family are selectively upregulated in tumours driven by C420R (Extended Data Fig. 6a). Analysis of The Cancer Genome Atlas (TCGA) data for glioma revealed that glypican 3 (*GPC3*) is more highly expressed in human glioblastoma tumours than in lower-grade glioma (Extended Data Fig. 6b). We confirmed protein expression using immunohistochemistry on primary human glioma samples and detected increased expression of GPC3 in C420R-driven mouse tumours (Fig. 5a, Extended Data Fig. 6c). Together, these data suggest that GPC3 may contribute to glioma tumorigenesis and downstream synaptic remodelling. Notably, how secreted synaptogenic proteins expressed in glioma influence (1) tumorigenesis, (2) the surrounding neuronal microenvironment and (3) key neurological features of glioma pathophysiology remain largely undefined. Therefore, examining the function of GPC3 in this context provides an opportunity to understand how these facets of glioma biology are regulated.

To determine whether GPC3 contributes to C420R-driven tumours, we used CRISPR–Cas9 approaches to delete GPC3 (GPC3^{cr}) in C420R-driven mouse tumours, and found that loss of GPC3 extended median survival (Fig. 5b, P44 versus P60, Extended Data Fig. 7). Notably, serial EEG recordings of mice bearing 2xCr; C420R; GPC3^{cr} tumours revealed a loss of early onset hyperexcitability compared to C420R-driven controls

(Fig. 5c). Having established that GPC3 is necessary for key pathophysiology properties of C420R-driven tumours, we next evaluated whether it is sufficient to drive glioma tumorigenesis independently of PIK3CA variants. We combined PiggyBac-mediated *Gpc3* overexpression with our 3xCr system; survival analysis revealed that mice overexpressing *Gpc3* exhibited significantly shorter median survival than 3xCr controls (P59 versus P91) (Fig. 5d, Extended Data Fig. 7). Next, we examined seizure phenotypes via video EEG in 3xCr tumours overexpressing *Gpc3*, and found that mice bearing *Gpc3*-overexpressing tumours exhibited early onset of convulsive seizures and heightened network hyperexcitability as early as P40 (Fig. 5d–f). Together, these studies indicate GPC3 functions as a driver of glioma tumorigenesis that accelerates the onset of seizures and hyperexcitability.

The robust effects of *Gpc3* overexpression on these key tumour pathophysiological properties led us to examine the cellular phenotypes in these tumours. First, we evaluated cellular proliferation, and found an increase in BrdU labelling in GPC3-driven tumours (Extended Data Fig. 8a). Next, we probed the peritumoral synaptic constituency, and identified increases in both excitatory and inhibitory synapses in these tumours (Extended Data Fig. 8b), which suggests that GPC3 promotes the aberrant formation of both excitatory and inhibitory synapses. Previous studies have demonstrated that other members of the glypican family promote synapse formation onto neurons through secretion from adjacent glial cells^{29,30}, suggesting that GPC3 may act by a similar mechanism. To test this, we overexpressed *Gpc3* in mouse astrocytes co-cultured with neurons and found that *Gpc3* overexpression in astrocytes markedly increases synapse formation onto neurons, as measured by expression of synaptic markers and whole-cell recordings from these neurons (Extended Data Fig. 9a–d). Conditioned medium from these astrocytes promoted heightened spontaneous postsynaptic current frequency on neurons (Extended Data Fig. 9e–g), which indicates that GPC3 acts via non-cell-autonomous mechanisms. These studies identify GPC3 as a key factor in

glioma-induced neosynaptogenesis that manipulates the neuronal microenvironment during tumour progression.

Discussion

Decoding variant function in cancer has become an increasingly crucial requirement for developing personalized therapeutic agents. In this study, we describe an *in vivo* complementation screening platform that enables the rapid identification of driver mutations in a native model of glioma. Using this system, we identified several driver variants of PIK3CA that are active in glioma. Crucially, a subset of these variants (C420R, M1043I and R88Q) appear to function in a glioma-relevant manner, as studies in other cancer models did not identify these variants, or their relative oncogenic potential varies^{7,17}. Together, these studies illustrate that variant function is context-specific and highlight the importance of decoding cell lineage and genetic relationships^{7,31–33}. Furthermore, among the PIK3CA variants that operate as drivers in our glioma system, transcriptome analysis identified marked molecular differences in the tumours driven by these variants. These findings demonstrate that similar variants can engender a diverse range of molecular properties in glioma tumours, reinforcing the importance of deciphering the selective functions of each driver variant.

Our identification of proliferative and synaptic gene dysregulation among tumours driven by H1047R and C420R served as an entry point for demonstrating phenotypic differences among these tumours. Using both native mouse and PDX models, we found that tumours driven by these variants selectively alter the neuronal microenvironment towards heightened network hyperexcitability and seizure onset—two key features of glioma pathophysiology²². Furthermore, H1047R and C420R use different mechanisms to engender these common physiological phenotypes. Nevertheless, these findings further illustrate that crosstalk between glioma tumours and the surrounding neuronal microenvironment is a key contributor to malignant growth. Several recent studies have shown that increases in neuronal activity promotes glioma tumour proliferation^{1,2}, and when put together with our findings, indicate a vicious cycle between glioma cells and adjacent neurons, in which glioma cells promote network hyperexcitability and these increases in neuronal activity promote tumour growth. To understand how these variant-driven tumours promote neuronal activity, we identified GPC3 as a driver of glioma tumorigenesis, network hyperexcitability and synapse formation. Because GPC3 promotes synapse formation through cell-non-autonomous mechanisms, a model emerges in which secreted proteins from glioma stimulates neuronal activity by triggering synaptogenesis during early tumorigenesis, working together with neurons to orchestrate both network hyperexcitability and growth.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-020-1952-2>.

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